The suspension of the two-point type is equivalent to the circle

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 588-591 and its proof environment opens on line 592, the next line; the pairing is by adjacency and the retained proof range 592-634 is the whole environment.

Editorial note. The book label-number dump gives this label as 6.5.1 (env lem, chapter 6, section 5, first counter value), so the counters chapter 6, section 5 and thm 0 (zero-based) are set so that the item prints with the book number.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command, and the transport facts used in the proof are standard prerequisites, not delegations of a step.

Editorial note. The author closes the second round trip with the sentence "so this follows easily", quoted here verbatim; the computation it refers to (the functoriality of the action on paths applied to the composite meridian) is written out in full on the preceding lines, so the main argument is complete.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the suspension of the two-point type is the circle (author label thm:suspbool, book Lemma 6.5.1).

Allowed standard prerequisites (editorial).

- Suspension and the circle. The suspension of a type has two poles and a meridian path for each point of the type, and the circle has a base point and a loop.. They are the two sides of the equivalence, and the maps are defined by their recursion principles.
- Recursion principles for the circle and the suspension. A map out of the circle is given by a point and a loop at it, and a map out of a suspension is given by two points together with a path between them for each point of the suspended type.. Both maps of the equivalence are introduced by these principles with explicit values on the generators.
- Transport in path spaces and pulled back fibrations. The transport of a path between the endpoints inside a family of path types is computed by conjugating with the action of the map on that path.. It turns the two

loop cases of the round-trip inductions into the explicit path equations the proof then simplifies.

Lemma 6.5.1. $\Sigma\mathbf{2} \simeq \mathbb{S}^1$.

Proof. Define $f : \Sigma\mathbf{2} \to \mathbb{S}^1$ by recursion such that $f(\mathsf{N}) :\equiv \mathsf{base}$ and $f(\mathsf{S}) :\equiv \mathsf{base}$, while $f(\mathsf{merid}(0_{\mathbf{2}})) \coloneqq \mathsf{loop}$ but $f(\mathsf{merid}(1_{\mathbf{2}})) \coloneqq \mathsf{refl}_{\mathsf{base}}$. Define $g : \mathbb{S}^1 \to \Sigma\mathbf{2}$ by recursion such that $g(\mathsf{base}) :\equiv \mathsf{N}$ and $g(\mathsf{loop}) \coloneqq \mathsf{merid}(0_{\mathbf{2}}) \cdot \mathsf{merid}(1_{\mathbf{2}})^{-1}$. We now show that f and g are quasi-inverses.

First we show by induction that $g(f(x)) = x$ for all $x : \Sigma\mathbf{2}$. If $x \equiv \mathsf{N}$, then $g(f(\mathsf{N})) \equiv g(\mathsf{base}) \equiv \mathsf{N}$, so we have $\mathsf{refl}_{\mathsf{N}} : g(f(\mathsf{N})) = \mathsf{N}$. If $x \equiv \mathsf{S}$, then $g(f(\mathsf{S})) \equiv g(\mathsf{base}) \equiv \mathsf{N}$, and we choose the equality $\mathsf{merid}(1_{\mathbf{2}}) : g(f(\mathsf{S})) = \mathsf{S}$. It remains to show that for any $y : \mathbf{2}$, these equalities are preserved as x varies along $\mathsf{merid}(y)$, which is to say that when $\mathsf{refl}_{\mathsf{N}}$ is transported along $\mathsf{merid}(y)$ it yields $\mathsf{merid}(1_{\mathbf{2}})$. By transport in path spaces and pulled back fibrations, this means we are to show that

$$g(f(\mathsf{merid}(y)))^{-1} \cdot \mathsf{refl}_{\mathsf{N}} \cdot \mathsf{merid}(y) = \mathsf{merid}(1_{\mathbf{2}}).$$

Of course, we may cancel $\mathsf{refl}_{\mathsf{N}}$. Now by $\mathbf{2}$-induction, we may assume either $y \equiv 0_{\mathbf{2}}$ or $y \equiv 1_{\mathbf{2}}$. If $y \equiv 0_{\mathbf{2}}$, then we have

$$\begin{aligned} g(f(\mathsf{merid}(0_{\mathbf{2}})))^{-1} \cdot \mathsf{merid}(0_{\mathbf{2}}) &= g(\mathsf{loop})^{-1} \cdot \mathsf{merid}(0_{\mathbf{2}}) \\ &= (\mathsf{merid}(0_{\mathbf{2}}) \cdot \mathsf{merid}(1_{\mathbf{2}})^{-1})^{-1} \cdot \mathsf{merid}(0_{\mathbf{2}}) \\ &= \mathsf{merid}(1_{\mathbf{2}}) \cdot \mathsf{merid}(0_{\mathbf{2}})^{-1} \cdot \mathsf{merid}(0_{\mathbf{2}}) \\ &= \mathsf{merid}(1_{\mathbf{2}}) \end{aligned}$$

while if $y \equiv 1_{\mathbf{2}}$, then we have

$$\begin{aligned} g(f(\mathsf{merid}(1_{\mathbf{2}})))^{-1} \cdot \mathsf{merid}(1_{\mathbf{2}}) &= g(\mathsf{refl}_{\mathsf{base}})^{-1} \cdot \mathsf{merid}(1_{\mathbf{2}}) \\ &= \mathsf{refl}_{\mathsf{N}}{}^{-1} \cdot \mathsf{merid}(1_{\mathbf{2}}) \\ &= \mathsf{merid}(1_{\mathbf{2}}). \end{aligned}$$

Thus, for all $x : \Sigma\mathbf{2}$, we have $g(f(x)) = x$.

Now we show by induction that $f(g(x)) = x$ for all $x : \mathbb{S}^1$. If $x \equiv \mathsf{base}$, then $f(g(\mathsf{base})) \equiv f(\mathsf{N}) \equiv \mathsf{base}$, so we have $\mathsf{refl}_{\mathsf{base}} : f(g(\mathsf{base})) = \mathsf{base}$. It remains to show that this equality is preserved as x varies along loop, which is to say that it is transported along loop to itself. Again, by transport in path spaces and pulled back fibrations, this means to show that

$$f(g(\mathsf{loop}))^{-1} \cdot \mathsf{refl}_{\mathsf{base}} \cdot \mathsf{loop} = \mathsf{refl}_{\mathsf{base}}.$$

However, we have

$$\begin{aligned} f(g(\mathsf{loop})) &= f\Big(\mathsf{merid}(0_{\mathbf{2}}) \cdot \mathsf{merid}(1_{\mathbf{2}})^{-1}\Big) \\ &= f(\mathsf{merid}(0_{\mathbf{2}})) \cdot f(\mathsf{merid}(1_{\mathbf{2}}))^{-1} \\ &= \mathsf{loop} \cdot \mathsf{refl}_{\mathsf{base}} \end{aligned}$$

so this follows easily. □